COMS4995W30
Applied Machine Learning

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From Models to Systems:
The Evolution of Applied Machine Learning

Course Journey at a Glance 🚀



What we covered (Weeks 1 → 11):

- ML problems → Metrics-driven learning
- Data & features → Linear models
- Decision trees → Ensembles
- Deep learning → Representation learning
- Sequence models → Transformers
- LLMs → RLHF and Agentic systems

Path: From supervised learning to LLM reasoning and agentic intelligence

What This Course Taught Us 🤔



How to formulate real-world problems as learning tasks

→ Define objectives, data requirements, and measurable success criteria

→ Translate business / scientific requirements into ML formulations

How to build end-to-end ML pipelines

→ Data cleaning, feature extraction, model training, evaluation, and deployment

→ Automate feedback loops and monitoring in production

What This Course Taught Us 🤔



How to design experiments and evaluate appropriately

→ Train/validation/test splits, cross-validation, ablation studies

→ Understand metrics: precision, recall, F1, AUCPR, perplexity

How to reason about trade-offs

→ Bias-variance, underfitting vs overfitting

→ Scaling vs cost (compute, labeling, iteration speed)

The Core ML Loop ♻️



The ML flywheel:

Data → Feature Engineering → Model → Loss → Optimization → Evaluation

- **The fundamental feedback loop underlying every ML system**
- A change in one stage can help or hurt the rest; evaluate the full system

The Core ML Loop ♻️



Practical Takeaways

- Always visualize intermediate results (train vs validation loss etc.)
- Apply regularization, dropout, or early stopping to balance fit and generalization
- Periodically re-evaluate when data distributions shift (“data drift”)

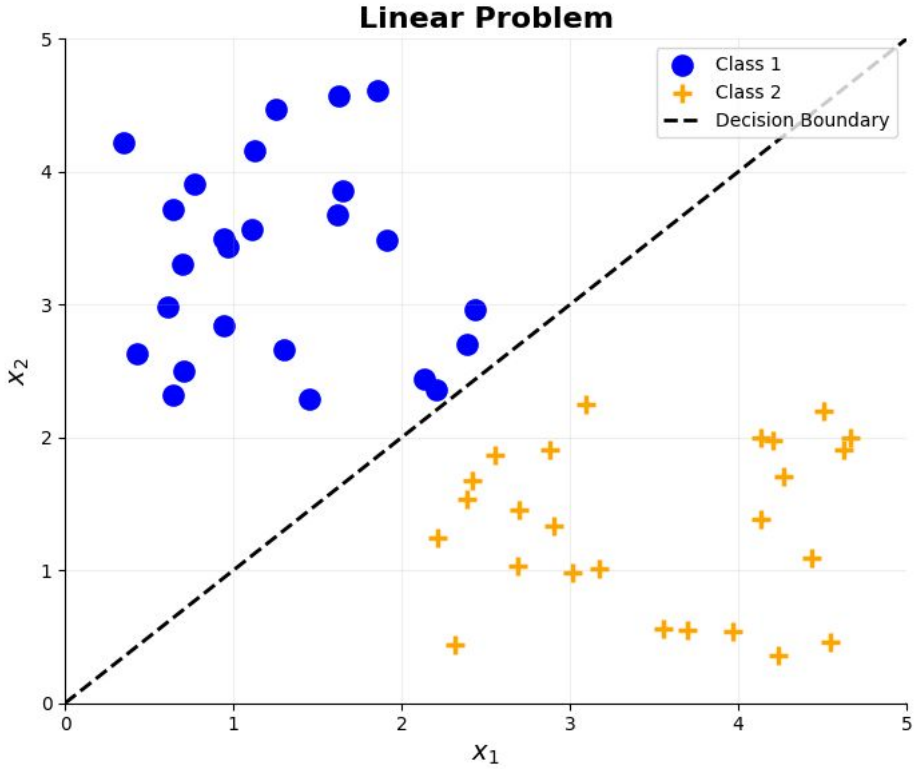
From Linear to Non-Linear Models

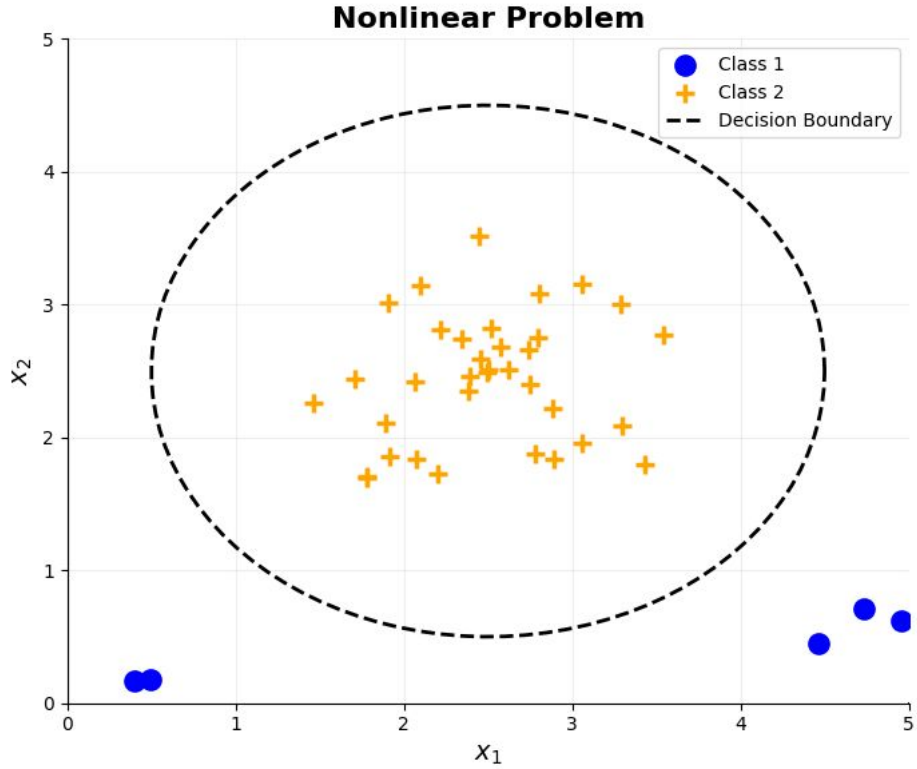


Linear/Logistic regression: simple, interpretable

Decision trees: handle non-linearity, feature interactions

Non-linear models unlock representation power *beyond linear boundaries*


Linear Problem
Class 1
Class 2
Decision Boundary
x_1
x_2


Nonlinear Problem
Class 1
Class 2
Decision Boundary
x_1
x_2

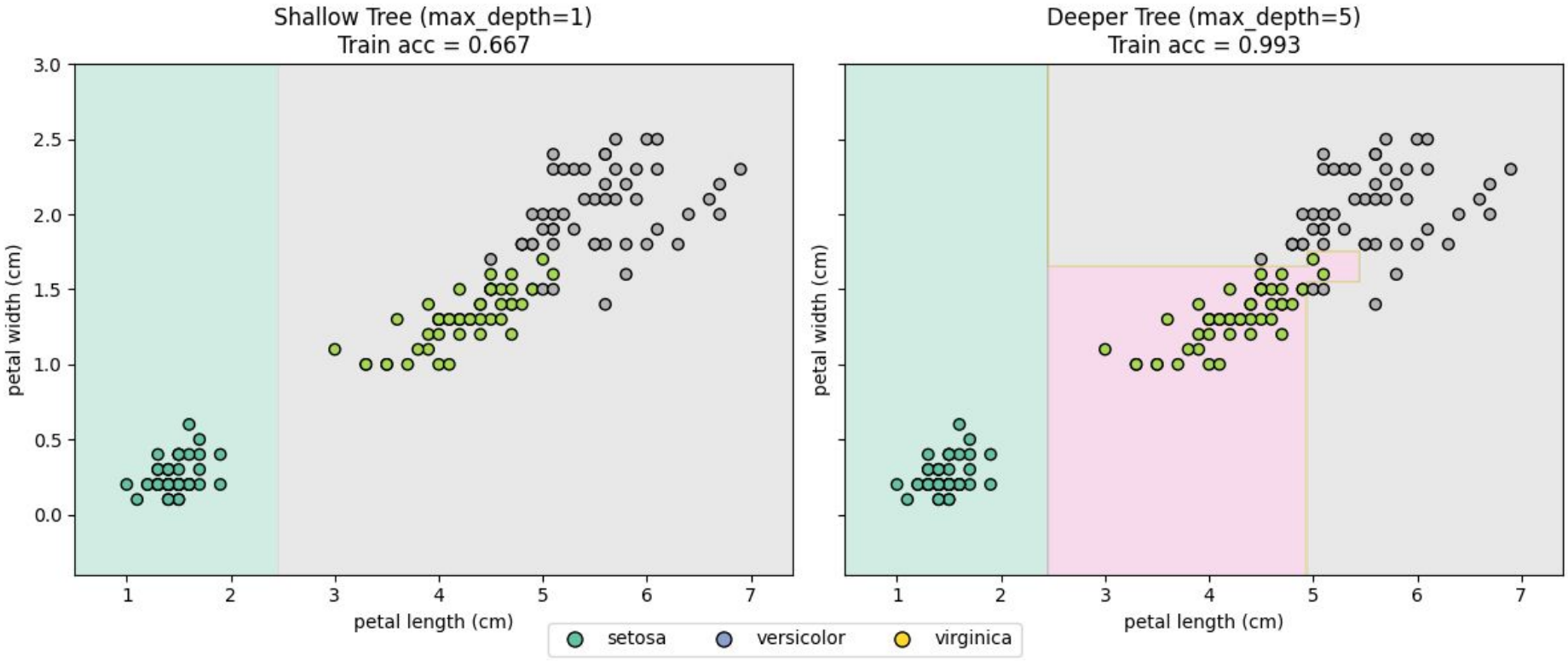
Ensembles: The Wisdom of Crowds

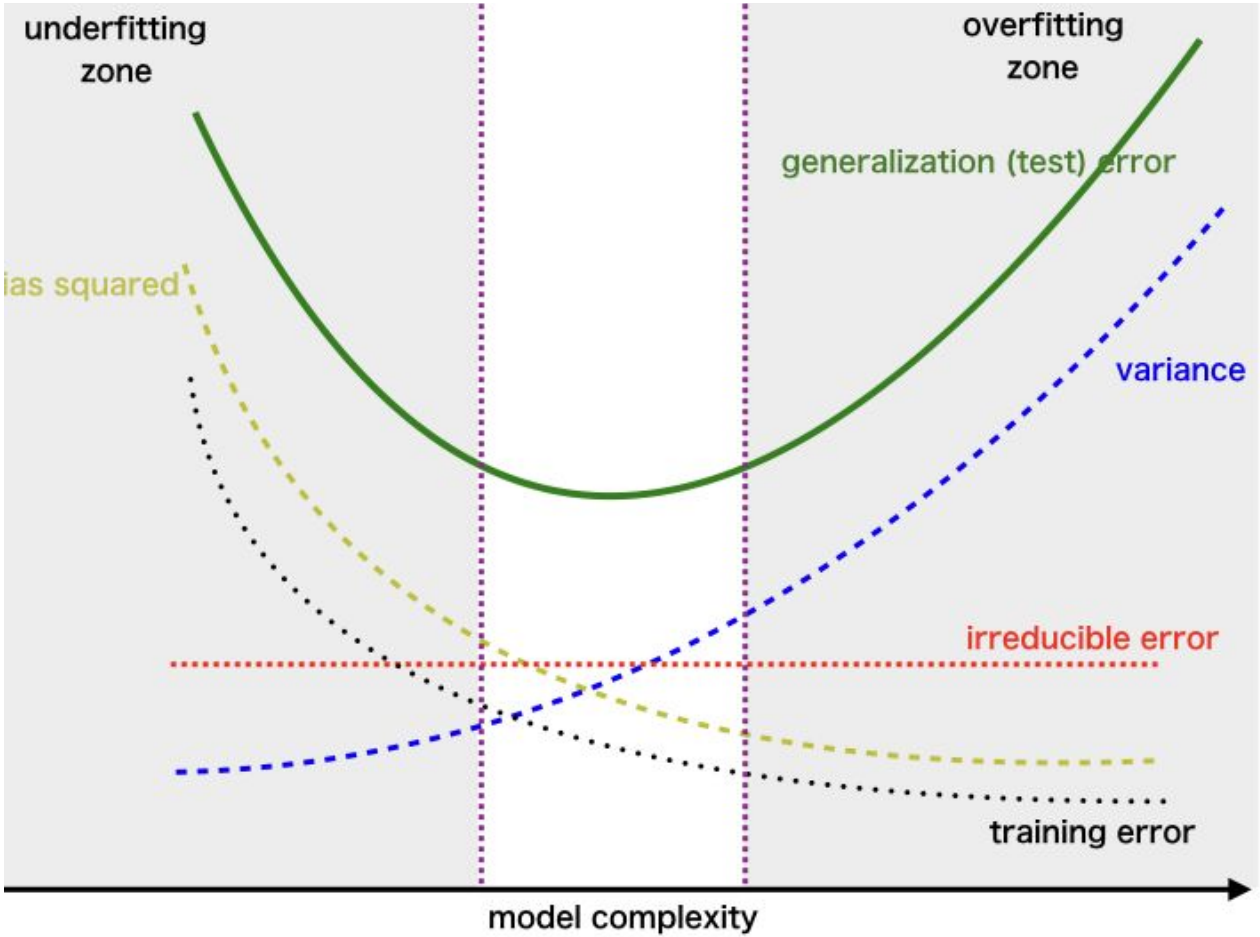


Bagging (Random Forest): reduce variance via many decorrelated trees

Boosting (XGBoost): sequentially correct errors, improve performance

Mixture-of-experts: conditional computation, with selected experts per token


Shallow Tree (max_depth=1)
Train acc = 0.667
Deeper Tree (max_depth=5)
Train acc = 0.993
petal width (cm)
petal length (cm)
setosa
versicolor
virginica


underfitting
zone
overfitting
zone
generalization (test) error
ias squared
variance
irreducible error
training error
model complexity

🎤 Mock Interview

A churn model scores 99% on a random split and 63% on next month's customers. The data includes "cancellation_date." What comes first?

A) Add a deeper neural network
B) Audit leakage and build a time-based evaluation split
C) Tune on next month's labels
D) Average more random splits

3 minutes with a partner. Choose one answer and explain why.

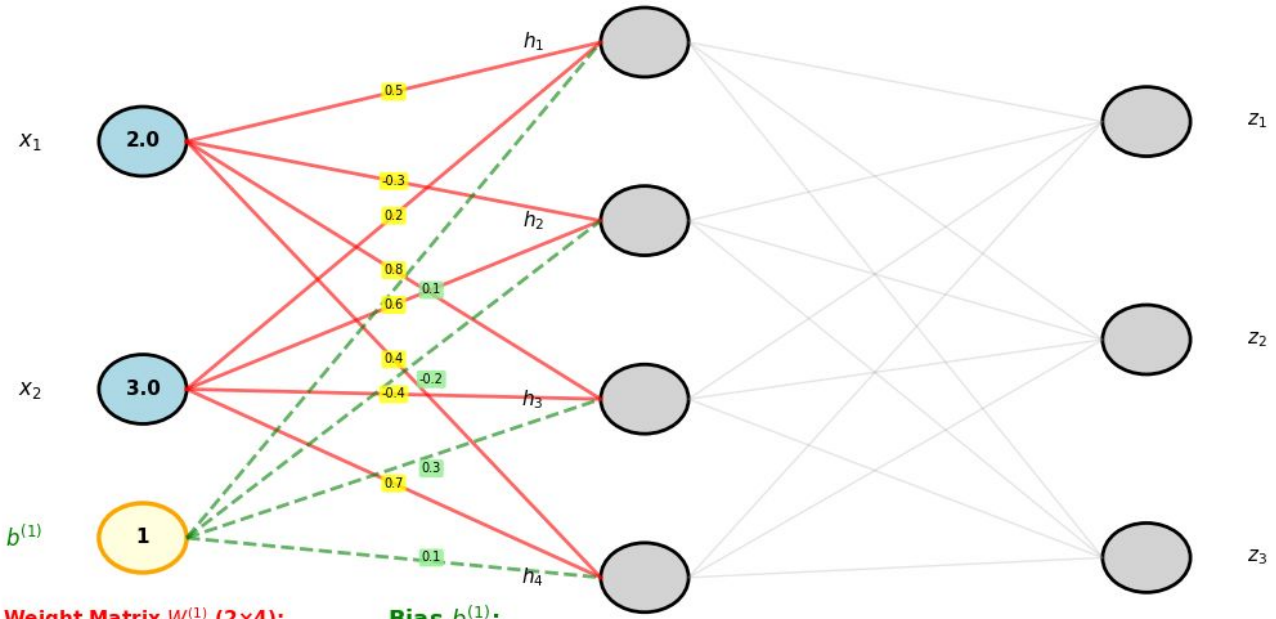
Deep Learning: Representation Learning



Forward pass

Loss computation

Backpropagation

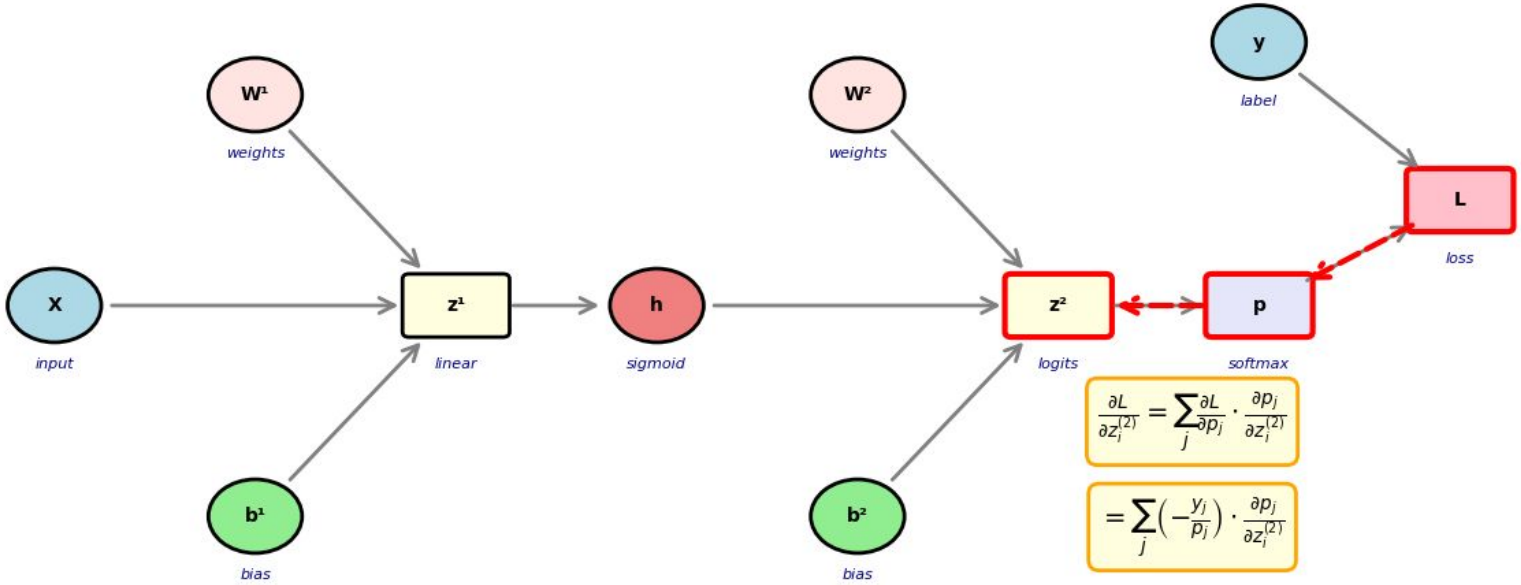


Weight Matrix $W^{(1)}$ (2×4):

$$\begin{bmatrix} 0.5 & -0.3 & 0.8 & 0.4 \\ 0.2 & 0.6 & -0.4 & 0.7 \end{bmatrix}$$

Bias $b^{(1)}$:

[0.1, -0.2, 0.3, 0.1]



$$\frac{\partial L}{\partial z_i^{(2)}} = \sum_j \frac{\partial L}{\partial p_j} \cdot \frac{\partial p_j}{\partial z_i^{(2)}}$$

$$= \sum_j \left(-\frac{y_j}{p_j} \right) \cdot \frac{\partial p_j}{\partial z_i^{(2)}}$$

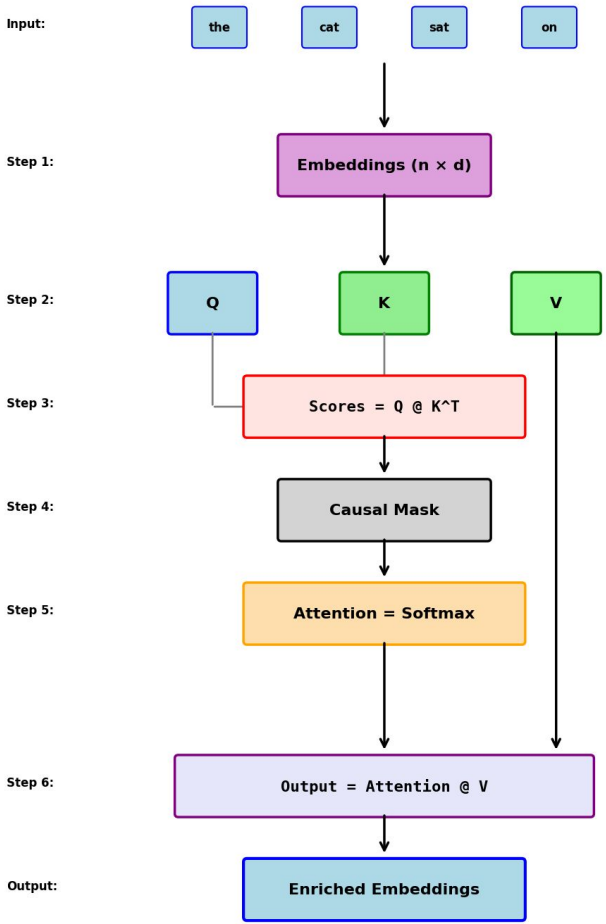
Softmax derivative:

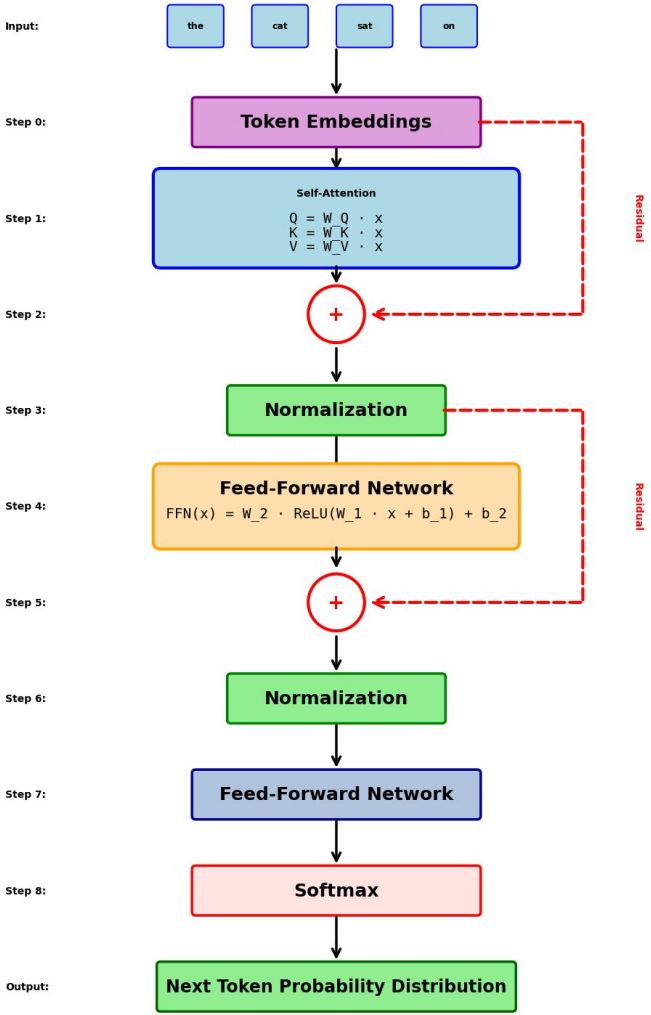
if $i = j$: $\frac{\partial p_i}{\partial z_i} = p_i(1 - p_i)$

if $i \neq j$: $\frac{\partial p_j}{\partial z_i} = -p_i p_j$

Final result:

$$\frac{\partial L}{\partial z_i^{(2)}} = p_i - y_i$$

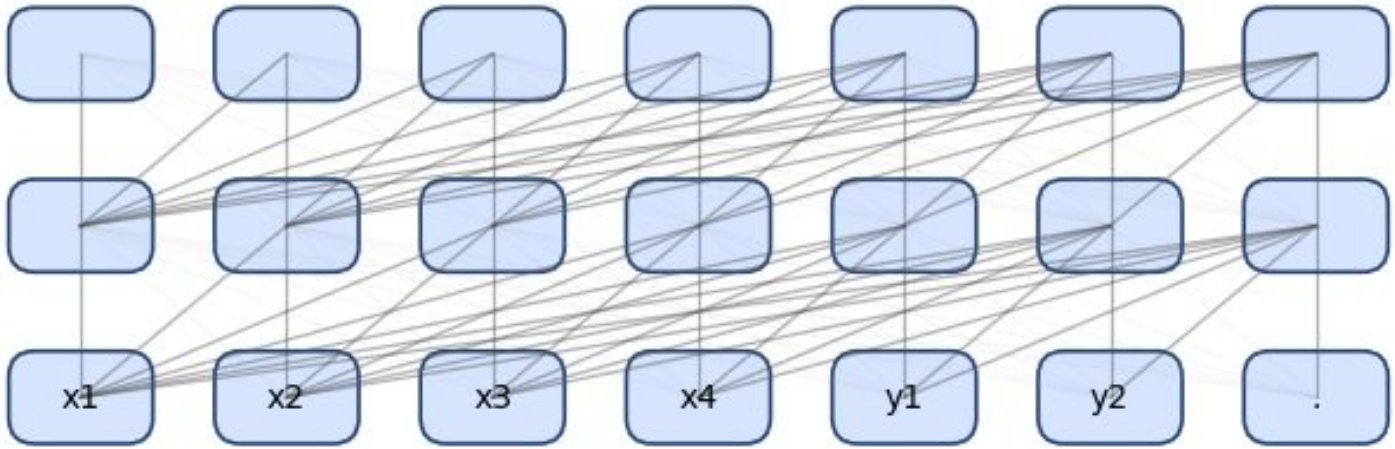

Input:
the
cat
sat
on
Step 1:
Embeddings (n × d)
Step 2:
Q
K
V
Step 3:
Scores = Q @ K^T
Step 4:
Causal Mask
Step 5:
Attention = Softmax
Step 6:
Output = Attention @ V
Output:
Enriched Embeddings

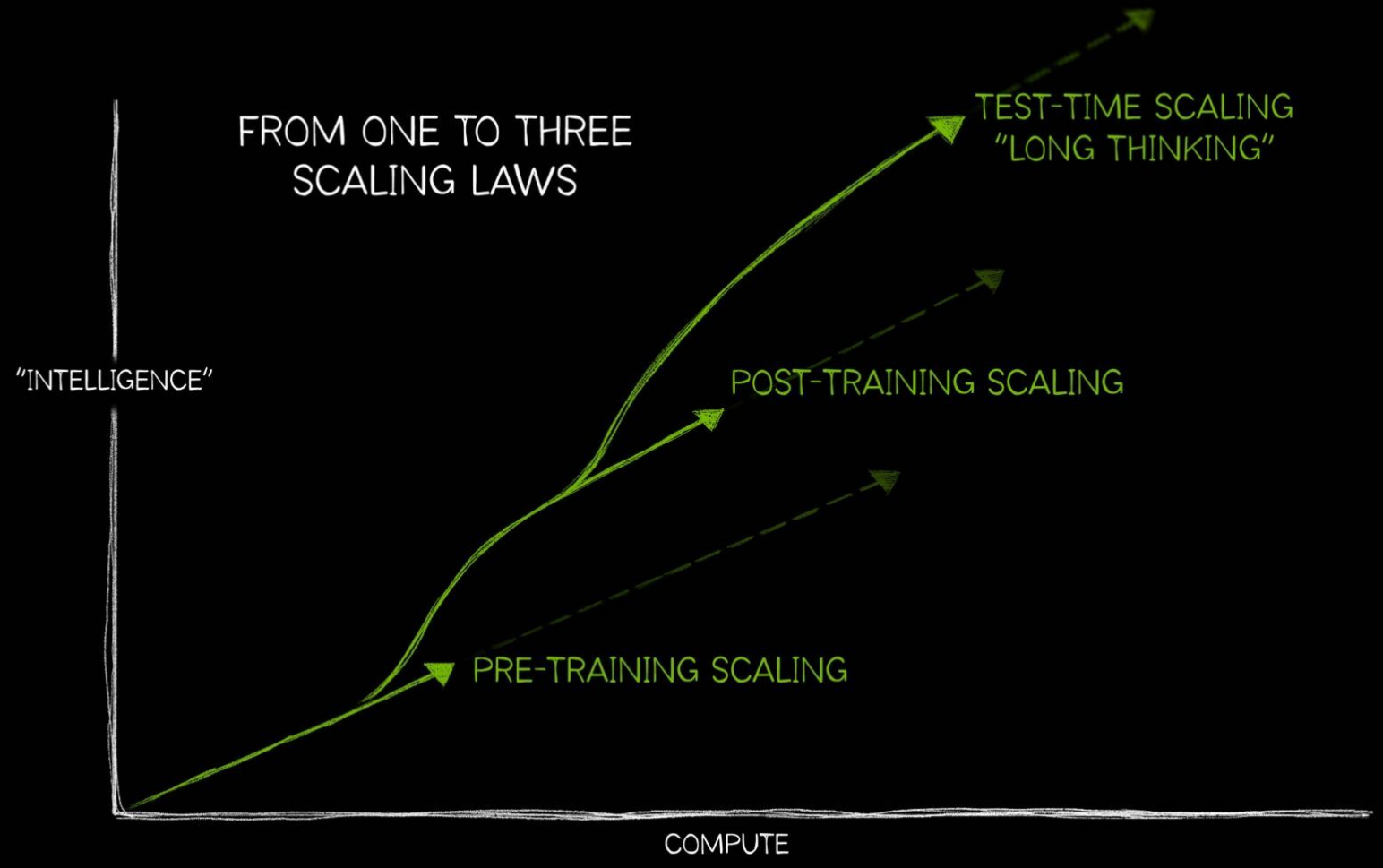


🤖 Attention is All You Need

Decoder-only

causal self-attention




FROM ONE TO THREE
SCALING LAWS
TEST-TIME SCALING
"LONG THINKING"
POST-TRAINING SCALING
PRE-TRAINING SCALING
"INTELLIGENCE"
COMPUTE

Pre-training on Everything

Self-supervised pre-training:

- Train on massive text corpora by predicting masked or next tokens

What it learns:

- Syntax, semantics, reasoning patterns, and broad world knowledge

Outcome:

- General-purpose representations adaptable to many tasks

Impact:

- Pre-training builds the foundation; fine-tuning shapes the purpose

Review: Which SFT Tokens Count?



A training example contains instructions, policy facts and an assistant response. What changes when only response tokens contribute to the loss?

DISCUSS, THEN CHECK

The prompt remains visible as conditioning context.

The selected response positions contribute to the objective.

The model is trained to produce the target response.

Alignment through RLxF



RLxF = Reinforcement Learning from X Feedback

- Uses human preference + reward models + verifiable rewards to guide behavior

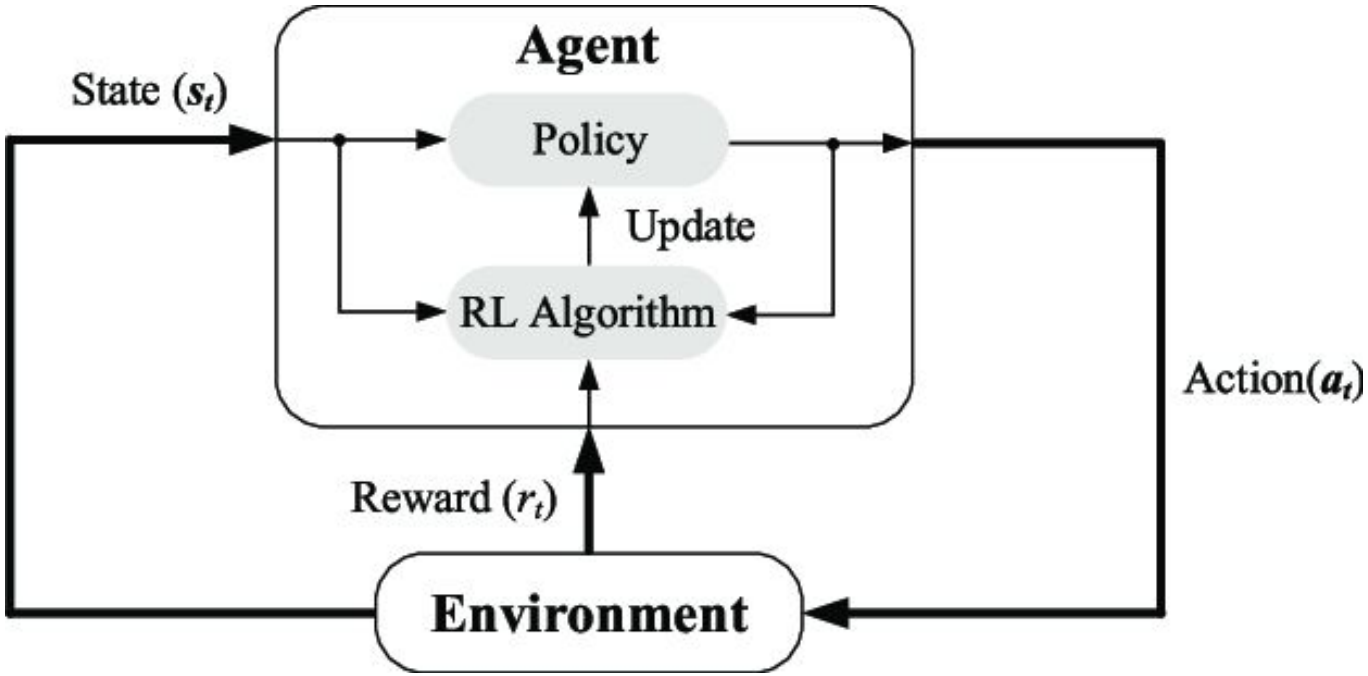
Process:

1️⃣ Humans/LLMs rank outputs

2️⃣ Use a learned preference reward or an external verifier

3️⃣ LLM optimized via PPO to maximize those signals

Goal: Align models with verifiable/human preference signals 🏅


Agent
State (s_t)
Policy
Update
RL Algorithm
Action(a_t)
Reward (r_t)
Environment

PPO: Proximal Policy Optimization 🧩



Core idea: small, controlled policy updates

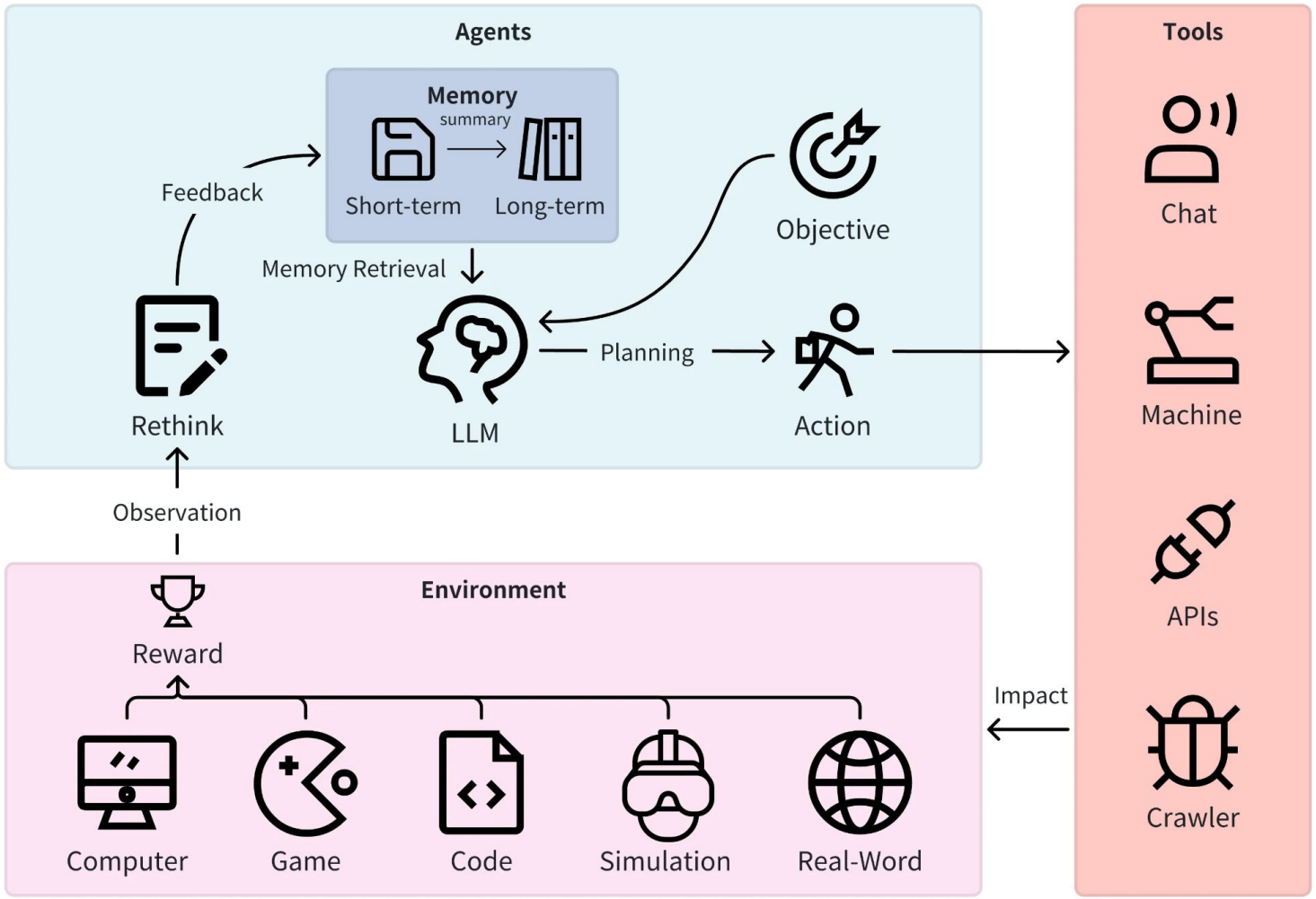
Maximize the clipped surrogate objective:

$$L_{clip} = E[\min(r A, \text{clip}(r, 1 - \varepsilon, 1 + \varepsilon) A)]$$

$$r = \pi_{new}(a \mid s) / \pi_{old}(a \mid s)$$

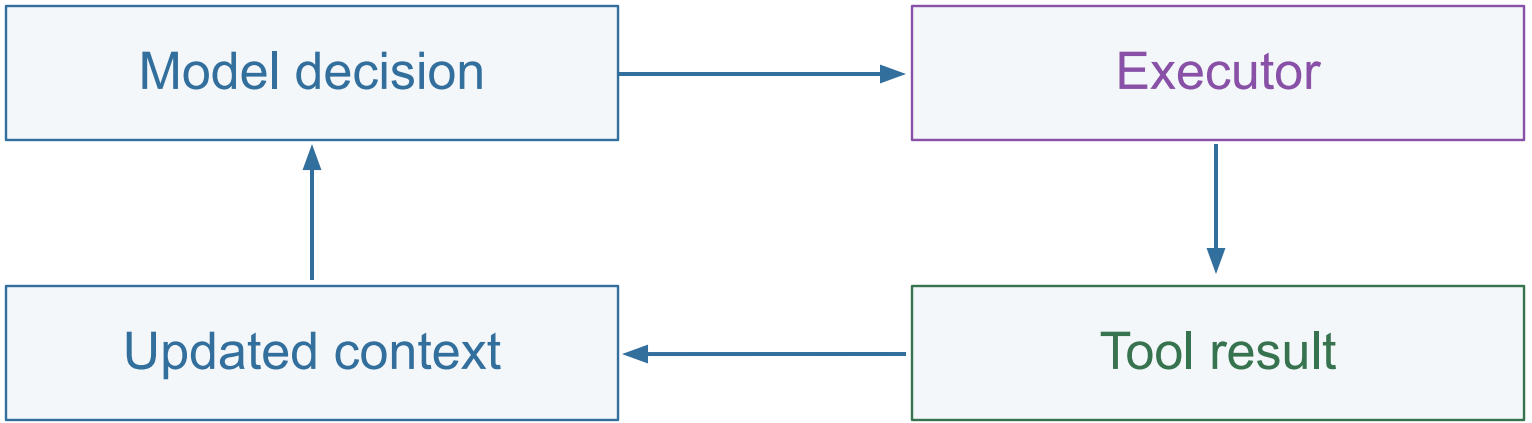
For LLM alignment, a KL penalty can also discourage drift from a fixed reference model.

Classic paper: Schulman et al., PPO (2017)


Agents
Memory
summary
Short-term
Long-term
Feedback
Objective
Memory Retrieval
Rethink
LLM
Planning
Action
Tools
Chat
Machine
APIs
Crawler
Observation
Environment
Reward
Computer
Game
Code
Simulation
Real-Word
Impact

Review: The Agent Loop

The executor turns a requested action into an observation.



What is Next?



Smaller, faster, cheaper LLMs with competitive performance

Agentic application everywhere

Multimodality: integrating text, vision, audio, action…

Advanced alignment: RLHF → RLAIF → Verifiable Rewards → 🤔

New architectures beyond Transformers

Better Agentic framework to make our lives even better

Skills to Build Beyond This Course



Short term (to landing a job): Strong Python/C++ (vibe debugging/coding), solid ML fundamentals, hands-on project experience, and the ability to clearly COMMUNICATE ideas and work with others

Long term (to stay competitive): Develop rapid learning skills, the ability to build and debug quickly with AI, and cross-domain understanding to leverage AI for fast prototyping and real-world impact - then share your lessons to establish your brand and reputation

Mindset: Aim to become a full-stack problem solver - someone who can use AI tools to build, analyze, and deploy ideas **FASTER** 🚀 than before, and learn from success and failures

🎤 Mock Interview



A policy assistant needs this week's rules, a fixed JSON format, and permission to issue refunds. Which design best fits?

A) Pre-train a larger model for all three

B) Use retrieval for rules, validate the output schema, and gate refund tools

C) Fine-tune once and remove the policy database

D) Let retrieved documents grant tool permissions

3 minutes with a partner. Choose one answer and explain why.

Explore the ML Space

as vast as the stars and the sea

